Chapter 2: Algorithm Discovery and Design

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Objectives

In this chapter, you will learn about

- Representing algorithms
- Examples of Algorithmic Problem Solving

Representing Algorithms

- Natural language
  - Language spoken and written in everyday life
  - Problems with using natural language for algorithms
    - Verbose
    - Imprecise
      - Relies on context and experiences to give precise meaning to a word or phase

Adding Two m-Digit Numbers

FIGURE 2.1

The Addition Algorithm of Figure 1.2 Expressed in Natural Language

Initially, set the value of the variable *carry* to 0 and the value of the variable i to 0. When these initializations have been completed, begin looping as long as the value of the variable i is less than or equal to $(m - 1)$. First, add together the values of the two digits a_i and b_i and the current value of the carry digit to get the result called c_i. Now check the value of c_i to see whether it is greater than or equal to 10. If c_i is greater than or equal to 10, then reset the value of *carry* to 1 and reduce the value of c_i by 10; otherwise, set the value of *carry* to zero. When you are done with that operation, add 1 to i and begin the loop all over again. When the loop has completed execution, set the leftmost digit of the result c_m to the value of *carry* and print out the final result, which consists of the digits $c_m\ c_m{-}1\ \ldots\ c_0$. After printing the result, the algorithm is finished, and it terminates.

Figure 2.1
The Addition Algorithm of Figure 1.2 Expressed in Natural Language

Representing Algorithms (continued)

- High-level programming language
  - Examples: C++, Java, Python
  - Problem with using a high-level programming language for algorithms
    - During the initial phases of design, we are forced to deal with detailed language issues

```
{
int i, m, Carry;
int[] a = new int[100];
int[] b = new int[100];
int[] c = new int[100];
m = Console.readInt();
for (int j = 0; j < =m-1; j++)   {
        a[j] = Console.readInt();
        b[j] = Console.readInt();
}
Carry = 0;
i = 0;
while (i < m)   {
        c[i] = a[i] + b[i] + Carry;
        if (c[i] > =10)
        .
        .
        .
```

Figure 2.2
The Beginning of the Addition Algorithm of Figure 1.2 Expressed in a High-Level Programming Language

Algorithm for Adding Two m-Digit Numbers

Given: $m \geq 1$ and two positive numbers each containing m digits, $a_{m-1}\, a_{m-2}, \ldots a_0$ and $b_{m-1}\, b_{m-2}, \ldots b_0$

Wanted: $c_m c_{m-1}\, c_{m-2} \ldots c_0$, where $c_m c_{m-1}\, c_{m-2} \ldots c_0 = (a_{m-1}\, a_{m-2} \ldots a_0) + (b_{m-1}\, b_{m-2} \ldots b_0)$

Algorithm:

Step 1 Set the value of *carry* to 0.

Step 2 Set the value of i to 0.

Step 3 While the value of i is less than or equal to $m - 1$, repeat the instructions in steps 4 through 6.

Step 4 Add the two digits a_i and b_i to the current value of *carry* to get c_i.

Step 5 If $c_i \geq 10$, then reset c_i to $(c_i - 10)$ and reset the value of *carry* to 1; otherwise, set the new value of *carry* to 0.

Step 6 Add 1 to i, effectively moving one column to the left.

Step 7 Set c_m to the value of *carry*.

Step 8 Print out the final answer, $c_m\, c_{m-1}\, c_{m-2} \ldots c_0$.

Step 9 Stop.

Figure 1.2
Algorithm for Adding Two m-digit Numbers

Pseudocode

- English language constructs modeled to look like statements available in most programming languages
- Steps presented in a structured manner (numbered, indented, and so on)
- No fixed syntax for most operations is required

Pseudocode (continued)

- Less ambiguous and more readable than natural language
- Emphasis is on process, not notation
- Well-understood forms allow logical reasoning about algorithm behavior
- Can be easily translated into a programming language

Sequential Operations

- Computation operations
  - Example
    - Set the value “variable” to “arithmetic expression”
    - Set the value “carry” to “0”
  - Variable
    - Named storage location that can hold a data value

- Set the value “carry” to “0”
- Set the value “carry” to “4+5”
- Set the value “carry” to “carry + 1”

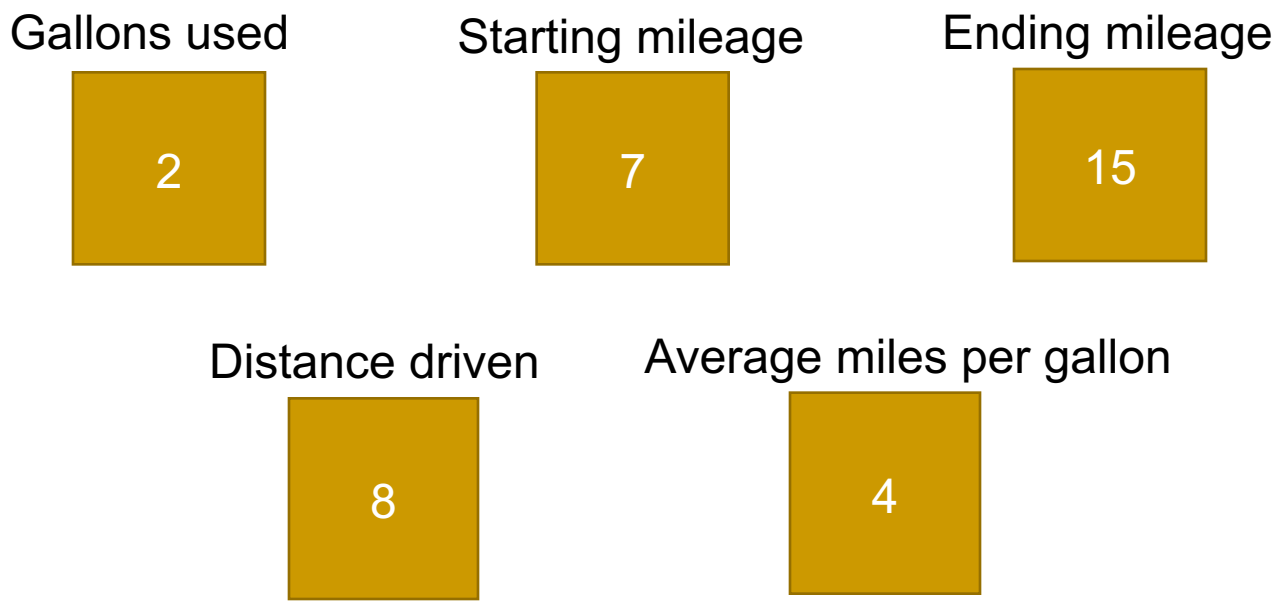
carry

Sequential Operations (continued)

- Input operations
  - To receive data values from the outside world
  - Example
    - Get a value for *r*, the radius of the circle
- Output operations
  - To send results to the outside world for display
  - Example
    - Print the value of *Area*

Average Miles per Gallon Algorithm (Version 1)

Step	Operation
1	Get values for *gallons used*, *starting mileage*, *ending mileage*
2	Set value of *distance driven* to (*ending mileage* – *starting mileage*)
3	Set value of *average miles per gallon* to (*distance driven* ÷ *gallons used*)
4	Print the value of *average miles per gallon*
5	Stop



Conditional and Iterative Operations

- Sequential algorithm
  - Also called straight-line algorithm
  - Executes its instructions in a straight line from top to bottom and then stops
- Control operations
  - Conditional operations
  - Iterative operations

Conditional and Iterative Operations (continued)

- Conditional operations
  - Ask questions and choose alternative actions based on the answers
  - Example
    - if *x* is greater than 25 then

      print *x*

      else

      print *x* times 100

FIGURE 2.4

Second Version of the Average Miles per Gallon Algorithm

Average Miles per Gallon Algorithm (Version 2)

STEP	OPERATION
1	Get values for *gallons used*, *starting mileage*, *ending mileage*
2	Set value of *distance driven* to (*ending mileage* – *starting mileage*)
3	Set value of *average miles per gallon* to (*distance driven* ÷ *gallons used*)
4	Print the value of *average miles per gallon*
5	If *average miles per gallon* is greater than 25.0 then
6	Print the message 'You are getting good gas mileage'
	Else
7	Print the message 'You are NOT getting good gas mileage'
8	Stop

Figure 2.5
Second Version of the Average Miles per Gallon Algorithm

Algorithm for average of three numbers

response=yes

While ***response*** = yes do

Get the values of x,y,z

Set the value of sum to (x+y+z)

If sum is greater than 0

Set the value of average to sum/3

Print the average

Else

Print “bad data”

Print the message “Do you want to do again”?

Get the value of ***response*** from the user

End loop

Stop

Conditional and Iterative Operations (continued)

- Iterative operations – **while** statement
  - Perform “looping” behavior, repeating actions until a continuation condition becomes false
  - Loop
    - The repetition of a block of instructions

Conditional and Iterative Operations (continued)

- Examples
  - while $j > 0$ do

    set s to $s + a_j$

    set j to $j - 1$

  - repeat

    print a_k

    set k to $k + 1$

    until $k > n$

Conditional and Iterative Operations (continued)

- **Components of a loop**
  - Continuation condition
  - Loop body
- **Infinite loop**
  - The continuation condition never becomes false
  - An error!

Average Miles per Gallon Algorithm (Version 3)

STEP	OPERATION
1	*response* = Yes
2	While (*response* = Yes) do steps 3 through 11
3	Get values for *gallons used*, *starting mileage*, *ending mileage*
4	Set value of *distance driven* to (*ending mileage* – *starting mileage*)
5	Set value of *average miles per gallon* to (*distance driven* ÷ *gallons used*)
6	Print the value of *average miles per gallon*
7	If average miles per gallon > 25.0 then
8	Print the message 'You are getting good gas mileage'
	Else
9	Print the message 'You are NOT getting good gas mileage'
10	Print the message 'Do you want to do this again? Enter Yes or No'
11	Get a new value for *response* from the user
12	Stop

Figure 2.7
Third Version of the Average Miles per Gallon Algorithm

Conditional and Iterative Operations (continued)

- **Pretest loop**
  - Continuation condition tested at the beginning of each pass through the loop
  - It is possible for the loop body to never be executed
  - → **While loop**

Conditional and Iterative Operations (continued)

- **Posttest loop**
  - Continuation condition tested at the end of loop body
  - Loop body must be executed at least once
  - → **Do/While loop**

FIGURE 2.6

Summary of Pseudocode Language Instructions

Figure 2.9

Summary of Pseudocode Language Instructions

COMPUTATION:

Set the value of "variable" to "arithmetic expression"

INPUT/OUTPUT:

Get a value for "variable", "variable"...
Print the value of "variable", "variable", ...
Print the message 'message'

CONDITIONAL:

```
If "a true/false condition" is true then
    first set of algorithmic operations
Else
    second set of algorithmic operations
```

ITERATIVE:

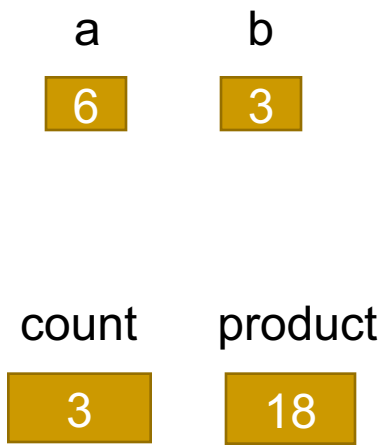
```
While ("a true/false condition") do step i through step j
        Step i: operation
        .
        .
        .
        Step j: operation

While ("a true/false condition") do
        operation
        .
        .
        .
        operation
End of the loop

Do
        operation
        operation
        .
        .
        .
While ("a true/false condition")
```

Examples of Algorithmic Problem Solving

- Go Forth and Multiply
  - Multiply two numbers using repeated addition
- Sequential search
  - Find a particular value in an unordered collection
- Find maximum
  - Find the largest value in a collection of data
- Pattern matching
  - Determine if and where a particular pattern occurs in a piece of text

Example 1: Go Forth and Multiply

- Task
  - Implement an algorithm to multiply two numbers, *a* and *b*, using **repeated addition**
- Algorithm outline
  - Create a loop that executes exactly *b* times, with each execution of the loop adding the value of *a* to a running total



Multiplication via Repeated Addition

```
Get values for a and b
If (either a = 0 or b = 0) then
    Set the value of product to 0
Else
    Set the value of count to 0
    Set the value of product to 0
    While (count < b) do
        Set the value of product to (product + a)
        Set the value of count to (count+1)
    End of loop
Print the value of product
Stop
```

Figure 2.10
Algorithm for Multiplication via Repeated Addition

Example 2: Looking, Looking, Looking

- Task
  - Find a particular person's name from an **unordered list** of telephone subscribers
- Algorithm outline
  - Start with the first entry and check its name, then repeat the process for all entries

Example 2: Looking, Looking, Looking (continued)

- Algorithm discovery
  - Finding a solution to a given problem

Example 2: Looking, Looking, Looking (continued)

- Correct sequential search algorithm
  - Uses iteration to simplify the task
  - Refers to a value in the list using an index (or pointer)
  - Handles special cases (such as a name not found in the collection)
  - Uses the variable *Found* to exit the iteration as soon as a match is found

FIGURE 2.9

The Sequential Search Algorithm

Figure 2.13

The Sequential Search Algorithm

Sequential Search Algorithm

STEP	OPERATION
1	Get values for *NAME*, $N_1, \ldots, N_{10,000}$, and $T_1, \ldots, T_{10,000}$
2	Set the value of *i* to 1 and set the value of *Found* to NO
3	While both (*Found* = NO) and ($i \leq 10{,}000$) do steps 4 through 7
4	If *NAME* is equal to the *i*th name on the list N_i then
5	Print the telephone number of that person, T_i
6	Set the value of *Found* to YES
	Else (*NAME* is not equal to N_i)
7	Add 1 to the value of *i*
8	If (*Found* = NO) then
9	Print the message 'Sorry, this name is not in the directory'
10	Stop

i 5

NAME

Ahmet

N1	T1
N2	T2
N3	T3
.	
Ahmet	054332892
.	
N10000	T10000

Found

NO

Example 2: Looking, Looking, Looking (continued)

- The selection of an algorithm to solve a problem is greatly influenced by:

  the way the data for that problem is organized

Example 3: Big, Bigger, Biggest

- Task
  - Find the **largest value** from a list of values
- Algorithm outline
  - Keep track of the largest value seen so far (initialized to be the first in the list)
  - Compare each value to the largest seen so far, and keep the larger as the new largest

Example 3: Big, Bigger, Biggest (continued)

- Once an algorithm has been developed, it may itself be used in the construction of other, more complex algorithms
- **Library**
  - A collection of useful algorithms
  - An important tool in algorithm design and development

Example 3: Big, Bigger, Biggest (continued)

- Find Largest algorithm
  - Uses iteration and indices as in previous example
  - Updates *location* and *largest so far* when needed in the loop

FIGURE 2.10

Algorithm to Find the Largest Value in a List

Find Largest Algorithm

Get a value for n, the size of the list
Get values for $A_1, A_2, \ldots, A_n$, the list to be searched
Set the value of *largest so far* to A_1
Set the value of *location* to 1
Set the value of i to 2
While ($i \leq n$) do
    If $A_i >$ *largest so far* then
        Set *largest so far* to A_i
        Set *location* to i
    Add 1 to the value of i
End of the loop
Print out the values of *largest so far* and *location*
Stop

Figure 2.14
Algorithm to Find the Largest Value in a List

Example 4: Meeting Your Match

- Task
  - Find if and where a **pattern string** occurs within a longer piece of text
- Algorithm outline
  - Try each possible location of pattern string in turn
  - At each location, compare pattern characters against string characters

Example 4: Meeting Your Match (continued)

- **Abstraction**
  - Separating high-level view from low-level details
  - Key concept in computer science
  - Makes difficult problems intellectually manageable
  - Allows piece-by-piece development of algorithms

Example 4: Meeting Your Match (continued)

- **Top-down design**
  - When solving a complex problem
    - Create high-level operations in the first draft of an algorithm
    - After drafting the outline of the algorithm, return to the high-level operations and elaborate each one
    - Repeat until all operations are primitives

Example 4: Meeting Your Match (continued)

- **Pattern-matching algorithm**
  - Contains a loop within a loop
    - External loop iterates through possible locations of matches to pattern
    - Internal loop iterates through corresponding characters of pattern and string to evaluate match

Pattern-Matching Algorithm

```
Get values for n and m, the size of the text and the pattern, respectively
Get values for both the text T1 T2 ... Tn and the pattern P1 P2 ... Pm
Set k, the starting location for the attempted match, to 1
While (k ≤ (n - m +1)) do
    Set the value of i to 1
    Set the value of Mismatch to NO
    While both (i ≤ m) and (Mismatch = NO) do
        If Pi ≠ T(k+(i-1)) then
            Set Mismatch to YES
        Else
            Increment i by 1 (to move to the next character)
    End of the loop
    If Mismatch = NO then
        Print the message 'There is a match at position'
        Print the value of k
    Increment k by 1
End of the loop
Stop, we are finished
```

Figure 2.16

Final Draft of the Pattern-Matching Algorithm

Summary

- Algorithm design is a first step in developing an algorithm
- Algorithm design must
  - Ensure the **algorithm is correct**
  - Ensure the **algorithm is sufficiently efficient**
- Pseudocode is used to design and represent algorithms

Summary

- Pseudocode must be readable, unambiguous, and able to be analyzed
- Algorithm design is a creative process; it uses multiple drafts and top-down design to develop the best solution
- Abstraction is a key tool for good design